Derivation of the Fundamental Structure from Motion Equations

EE 243

1 Introduction and Notation Reference

In computer vision, the fundamental equations of motion relate the 3D velocity of a point in space to its corresponding 2D velocity (optical flow) on the image plane under a perspective projection model.

1.1 Notation Dictionary

Symbol	**Definition**
$\mathbf{P} = [X, Y, Z]^T$	3D coordinates of a point in the camera-centered coordinate frame
f	Focal length of the camera system
$\mathbf{x} = [x, y]^T$	2D coordinates of the projected point on the image plane
u, v	Components of optical flow (velocity on the image plane, equivalent to $\dot{x}, \dot{y}$)
$\mathbf{T} = [T_X, T_Y, T_Z]^T$	Linear translational velocity vector of the camera
$\boldsymbol{\omega} = [\omega_X, \omega_Y, \omega_Z]^T$	Angular rotational velocity vector of the camera

Table 1: Variables used in the Structure from Motion formulation.

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2 Step 1: 3D Rigid Body Kinematics

Let a 3D point in the scene relative to the camera coordinate frame be denoted by $\mathbf{P}$. If the camera moves through space with a translational velocity $\mathbf{T}$ and a rotational velocity $\boldsymbol{\omega}$, the relative velocity $\dot{\mathbf{P}}$ of the point with respect to the camera observer is given by the kinematic rigid body equation:

$$\dot{\mathbf{P}} = -\mathbf{T} - \boldsymbol{\omega} \times \mathbf{P} \tag{1}$$

Expanding the cross product vector $\boldsymbol{\omega} \times \mathbf{P}$:

$$\begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix} = - \begin{bmatrix} T_X \\ T_Y \\ T_Z \end{bmatrix} - \begin{bmatrix} \omega_Y Z - \omega_Z Y \\ \omega_Z X - \omega_X Z \\ \omega_X Y - \omega_Y X \end{bmatrix} \tag{2}$$

Breaking this down into individual scalar equations for each 3D coordinate axis yield:

$$\dot{X} = -T_X - \omega_Y Z + \omega_Z Y \tag{3}$$

$$\dot{Y} = -T_Y - \omega_Z X + \omega_X Z \tag{4}$$

$$\dot{Z} = -T_Z - \omega_X Y + \omega_Y X \tag{5}$$

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3 Step 2: Pinhole Perspective Projection Model

Under an ideal pinhole camera approximation, a 3D scene point projects onto the 2D image plane at location (x, y) according to the following perspective scaling relations:

$$x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z} \tag{6}$$

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4 Step 3: Temporal Differentiation (Image Coordinate Velocity)

To compute the optical flow coordinates $u = \dot{x}$ and $v = \dot{y}$, we must differentiate the perspective equations with respect to time using the calculus quotient rule.

4.1 Derivation for Horizontal Optical Flow (u)

Differentiating $x = f\frac{X}{Z}$ yields:

$$u = \dot{x} = \frac{d}{dt}\left(f\frac{X}{Z}\right) = f\frac{\dot{X}Z - X\dot{Z}}{Z^2} = \frac{f\dot{X}}{Z} - \frac{fX\dot{Z}}{Z^2} \tag{7}$$

We factor out the spatial component by utilizing the identity $X = \frac{xZ}{f}$:

$$u = \frac{f\dot{X} - x\dot{Z}}{Z} \tag{8}$$

Now, substituting the 3D velocity component equations (Equations 3 and 5) for $\dot{X}$ and $\dot{Z}$:

$$u = \frac{f\left(-T_X - \omega_Y Z + \omega_Z Y\right) - x\left(-T_Z - \omega_X Y + \omega_Y X\right)}{Z} \tag{9}$$

Distributing terms across the common denominator Z:

$$u = \frac{-fT_X + xT_Z}{Z} - f\omega_Y + f\omega_Z\left(\frac{Y}{Z}\right) + x\omega_X\left(\frac{Y}{Z}\right) - x\omega_Y\left(\frac{X}{Z}\right) \tag{10}$$

By reapplying the basic projection definitions ($\frac{Y}{Z} = \frac{y}{f}$ and $\frac{X}{Z} = \frac{x}{f}$), we can cleanly eliminate the unobservable 3D coordinates X and Y:

$$u = \frac{T_Z x - T_X f}{Z} + x\omega_X\left(\frac{y}{f}\right) - f\omega_Y - x\omega_Y\left(\frac{x}{f}\right) + f\omega_Z\left(\frac{y}{f}\right) \tag{11}$$

Factoring out the angular velocity components ω_X, ω_Y, and ω_Z reveals the final structured layout for the horizontal image plane tracking:

$$u = \frac{T_Z x - T_X f}{Z} + \omega_X\left(\frac{xy}{f}\right) - \omega_Y\left(f + \frac{x^2}{f}\right) + \omega_Z y \tag{12}$$

4.2 Derivation for Vertical Optical Flow (v)

Similarly, differentiating $y = f\frac{Y}{Z}$ with respect to time yields:

$$v = \dot{y} = \frac{d}{dt}\left(f\frac{Y}{Z}\right) = f\frac{\dot{Y}Z - Y\dot{Z}}{Z^2} = \frac{f\dot{Y}}{Z} - \frac{fY\dot{Z}}{Z^2} \tag{13}$$

Substituting the identity $Y = \frac{yZ}{f}$:

$$v = \frac{f\dot{Y} - y\dot{Z}}{Z} \tag{14}$$

Now, substituting the 3D velocity component equations (Equations 4 and 5) for $\dot{Y}$ and $\dot{Z}$:

$$v = \frac{f\left(-T_Y - \omega_Z X + \omega_X Z\right) - y\left(-T_Z - \omega_X Y + \omega_Y X\right)}{Z} \tag{15}$$

Distributing across the common denominator Z:

$$v = \frac{-fT_Y + yT_Z}{Z} - f\omega_Z\left(\frac{X}{Z}\right) + f\omega_X + y\omega_X\left(\frac{Y}{Z}\right) - y\omega_Y\left(\frac{X}{Z}\right) \tag{16}$$

Once more, transforming our 3D position fields back to known camera coordinates via ($\frac{Y}{Z} = \frac{y}{f}$ and $\frac{X}{Z} = \frac{x}{f}$):

$$v = \frac{T_Z y - T_Y f}{Z} + f\omega_X + y\omega_X\left(\frac{y}{f}\right) - y\omega_Y\left(\frac{x}{f}\right) - f\omega_Z\left(\frac{x}{f}\right) \tag{17}$$

Grouping the components systematically by their respective rotational terms:

$$v = \frac{T_Z y - T_Y f}{Z} + \omega_X\left(f + \frac{y^2}{f}\right) - \omega_Y\left(\frac{xy}{f}\right) - \omega_Z x \tag{18}$$

Note on Sign Conventions: Depending on the native handedness convention adopted for the frame rotation orientation (clockwise vs. counter-clockwise configurations around the focal primary optical axis), the system parameterizes the final rotational twist scalar directly as $+\omega_Z x$.

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5 Final Structure from Motion Summary Equations

Gathering our final results, we establish the standard matching framework for visual odometry and optical scene flow fields:

$$u = \frac{T_Z x - T_X f}{Z} + \omega_X\left(\frac{xy}{f}\right) - \omega_Y\left(f + \frac{x^2}{f}\right) + \omega_Z y \tag{19}$$

$$v = \frac{T_Z y - T_Y f}{Z} + \omega_X\left(f + \frac{y^2}{f}\right) - \omega_Y\left(\frac{xy}{f}\right) + \omega_Z x \tag{20}$$